

Assignment- 21. *Rabi oscillations*

- Calculate transition dipole moment for 1s-2p atomic transition in hydrogen atom for z-polarized light.
- Now consider a powerful beam of light is incident on gas of monoatomic hydrogen and is tuned to resonance with the 1s-2p transition at 137 nm.
 - Calculate the Rabi oscillation period when the optical intensity is 10kWm^{-2} and the light is polarized in the z-direction.
 - Calculate the optical intensity required to make the Rabi oscillation period equal to the radiative lifetime of the 2p level, which is 1.6 ns.

2. *Pulse area*: Typically observation of Rabi oscillation requires high-power laser beams (as you might have noted above), which is achieved by using pulsed lasers. In pulsed lasers electric field amplitude (E_0) varies with time, which causes Rabi frequency also to vary with time. Therefore, a new parameter, pulse area Θ , is defined as following:

$$\Theta = \left| \frac{\mu_{12}}{\hbar} \int_{-\infty}^{\infty} E_0(t) dt \right|$$

A pulsed laser beam is focused to a spot of radius $1\text{ }\mu\text{m}$ on a gas of atoms with a dipole moment 10^{-29}Cm at the laser frequency.

- Calculate the pulse energy required to rotate the Bloch vector by $\pi/2$ radians for Gaussian pulses with a duration (FWHM) of 1 ps.
 - If the system is initially in the ground state, find the state of the system at the end of the pulse.
3. *Density Matrices*:
- Prove $\text{Tr}(\hat{\rho}^2) = 1$ for a pure state and $\text{Tr}(\hat{\rho}^2) < 1$ for a mixed state.
 - Prove $\langle \hat{O} \rangle = \text{Tr}(\hat{\rho} \hat{O})$.
 - In a density matrix representation of a system, what condition between various density matrix elements will ensure maximum coherence?
4. *Density Matrices*: Consider an ensemble of two-level atoms with the two internal states $|g\rangle$ (ground state) and $|e\rangle$ (excited state).

- Assume that you have an ensemble of atoms that are all in the state $|\psi\rangle = \frac{\sqrt{3}}{2}|g\rangle + \frac{i}{2}|e\rangle$.

What is the density matrix of an atom in this ensemble in the basis $|g\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $|e\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$?

- Assume now that you have created an ensemble by mixing 75% ground state atoms and 25% excited state atoms. If you pick one atom at random out of this ensemble, what is its density matrix?

- Assume that somebody gave you an ensemble of atoms. How could you experimentally distinguish between the above two situation?

5. *Optical Bloch Equations*: As discussed in class, plot ρ_{22} as a function of time for (i) $\Omega_0 = 3\gamma$, $\delta = 0$, (ii) $\Omega_0 = 3\gamma$, $\delta = 0.1\gamma$, (iii) $\Omega_0 = \gamma$, $\delta = 0$, using a plotting tool like MATLAB.